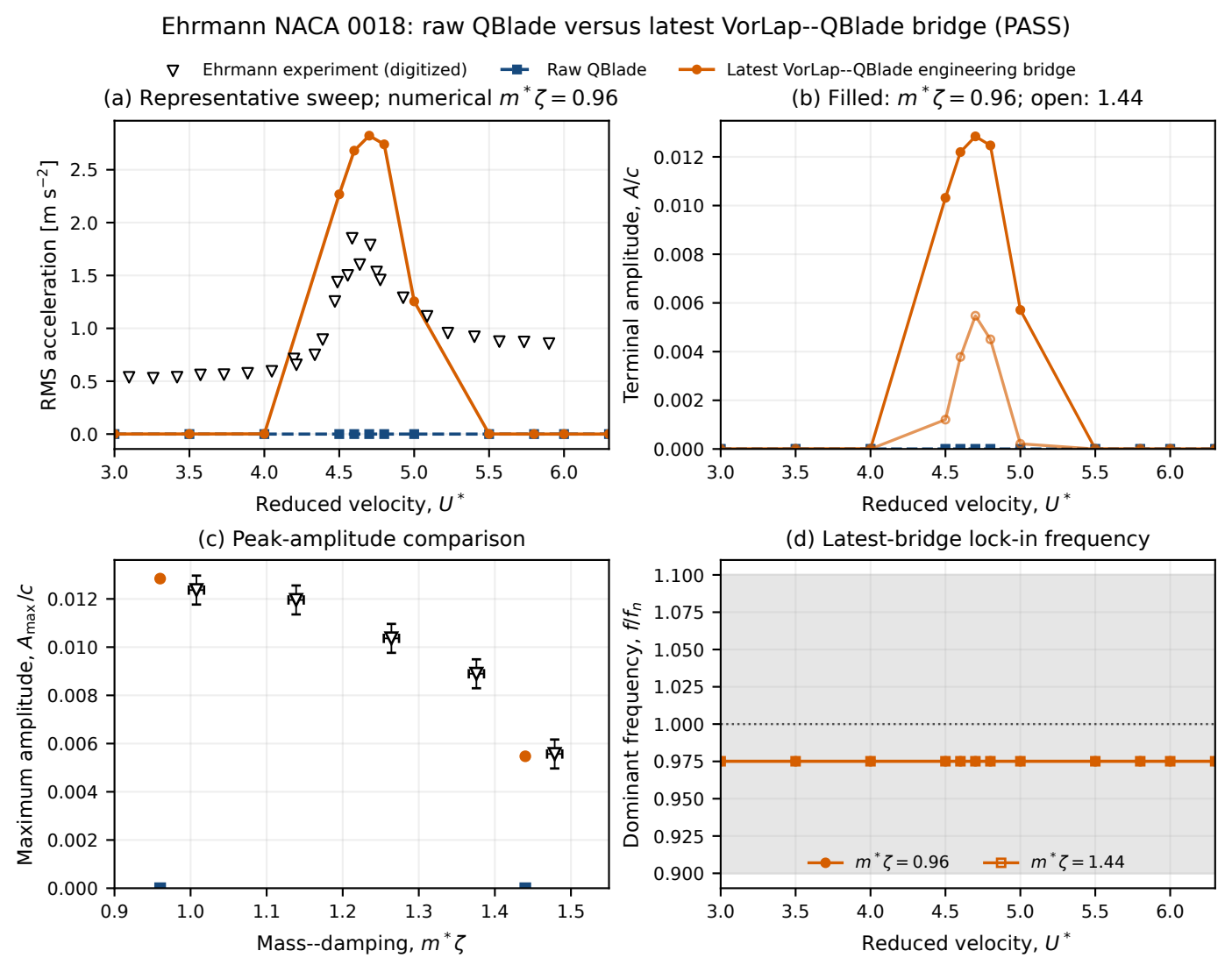

Ehrmann NACA 0018: raw QBlade versus latest VorLap--QBlade bridge (PASS)
Ehrmann experiment (digitized)
Raw QBlade
Latest VorLap--QBlade engineering bridge
(a) Representative sweep; numerical $m^*\zeta = 0.96$
RMS acceleration [m s^{-2}]
Reduced velocity, U^*
(b) Filled: $m^*\zeta = 0.96$; open: 1.44
Terminal amplitude, A/c
Reduced velocity, U^*
(c) Peak-amplitude comparison
Maximum amplitude, A_{max}/c
Mass--damping, $m^*\zeta$
(d) Latest-bridge lock-in frequency
Dominant frequency, f/f_n
$m^*\zeta = 0.96$
$m^*\zeta = 1.44$
Reduced velocity, U^*